

Aim: Demonstrate Data Validation

Code:

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8" />
<meta name="viewport"
content="width=device-width,initial-
scale=1" />
<title>Login / Signup</title>
</head>
<body>
<main class="container">
<section>
<div class="card">
<div class="tabs" role="tablist" aria-
label="Auth tabs">
<button
id="tab-login"
class="active"
role="tab"
aria-selected="true">
Login
</button>
<button
id="tab-signup"
role="tab"
aria-selected="false">
Sign up
</button>
</div>

<div id="login-panel">
```

```
<form id="login-form" novalidate>
<div>
<label for="login-identifier">Email or
Username</label>
<input
id="login-identifier"
name="identifier"
type="text"
required
placeholder="you@example.com or username"
/>
<div
class="error"
id="login-identifier-error"
></div>
</div>

<div>
<label for="login-
password">Password</label>
<div class="field-inline">
<input
id="login-password"
name="password"
type="password"
required
placeholder="Enter password"
/>
<button
type="button"
class="toggle-pass"
```

```

data-target="login-password">
Show
</button>
</div>
<div
class="error"
id="login-password-error"
></div>
</div>

<div class="actions">
<div class="small helper">
Forgot password? Click reset on server.
</div>
<button class="btn"
type="submit">Login</button>
</div>
<div
id="login-feedback"
class="muted"
role="status"
></div>
</form>
</div>

<div id="signup-panel" style="display:
none">
<form id="signup-form" novalidate>
<div>
<label for="username">Username</label>
<input
id="username"
name="username"
type="text"
required
placeholder="username (3-20 chars)"
/>
<div class="error" id="username-
error"></div>
</div>
<div>
<label for="email">Email</label>
<input
id="email"
name="email"
type="email"
required
placeholder="you@example.com"
/>
<div class="error" id="email-
error"></div>
</div>
<div class="row">
<div class="col">
<label for="phone">Phone</label>
<input
id="phone"
name="phone"
type="tel"
placeholder="+1234567890"
/>
<div class="error" id="phone-
error"></div>
</div>
<div class="col">

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<label for="dob">Date of birth</label>
<input
id="dob"
name="dob"
type="date"
required
/>
<div class="error" id="dob-error"></div>
</div>
</div>

<div>
<label for="password">Password</label>
<div class="field-inline">
<input
id="password"
name="password"
type="password"
required
placeholder="Minimum 8 characters"
/>
<button
type="button"
class="toggle-pass"
data-target="password">
Show
</button>
</div>
<div class="error" id="password-
error"></div>
</div>

<div>

```

```

<label for="confirm-password">Confirm
password</label>
<input
id="confirm-password"
name="confirmPassword"
type="password"
required
/>
<div
class="error"
id="confirm-password-error"
></div>
</div>

<div>
<label><input id="terms" type="checkbox"
/> I agree to the terms and privacy
policy</label>
<div class="error" id="terms-
error"></div>
</div>

<div class="actions">
<button class="btn" type="submit">
Create account
</button>
<button
type="button"
class="btn ghost"
id="fill-sample">
Fill sample
</button>
</div>

<div id="signup-feedback"
role="status"></div>

```

```

</form>
</div>
</div>
</section>
</main>

<script>

const qs = (s) =>
document.querySelector(s);

const qsa = (s) =>
document.querySelectorAll(s);

const tabLogin = qs("#tab-login");
const tabSignup = qs("#tab-signup");
const loginPanel = qs("#login-panel");
const signupPanel = qs("#signup-panel");

function showPanel(panel) {
  if (panel === "login") {
    tabLogin.classList.add("active");
    tabLogin.setAttribute("aria-selected",
      "true");
    tabSignup.classList.remove("active");
    tabSignup.setAttribute("aria-selected",
      "false");
    loginPanel.style.display = "block";
    signupPanel.style.display = "none";
  } else {
    tabSignup.classList.add("active");
    tabSignup.setAttribute("aria-selected",
      "true");
    tabLogin.classList.remove("active");
    tabLogin.setAttribute("aria-selected",
      "false");
    loginPanel.style.display = "none";
    signupPanel.style.display = "block";
  }
}

const validators = {
  username: (v) =>
    /^[A-Za-z0-9_]{3,20}$/ .test(v) ||
    "3-20 chars; letters, numbers, underscore
    only",
  email: (v) =>
    /^[^\s@]+@[^\s@]+\.[^\s@]+$/ .test(v) ||
    "Enter a valid email",
  phone: (v) =>
    v.trim() === "" ||
    /^[+]?[0-9]{10,15}$/ .test(v) ||
    "Enter 10-15 digits, optional +",
  password: (v) =>

```

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(v.length >= 8 &&
/[a-z]/.test(v) &&
/[A-Z]/.test(v) &&
/\d/.test(v)) ||
"Min 8 chars with upper, lower and
number",
confirmPassword: (v, all) =>
v === all.password || "Passwords do not
match",
dob: (v) => {
if (!v) return "Date of birth is
required";
const dob = new Date(v);
if (Number.isNaN(dob)) return "Invalid
date";
const today = new Date();
const age =
today.getFullYear() -
dob.getFullYear() -
(today <
new Date(
dob.getFullYear() +
(today.getFullYear() -
dob.getFullYear()),
dob.getMonth(),
dob.getDate(),
)
? 1
: 0);
return age >= 13 || "You must be at least
13 years old";
},
terms: (v) => v === true || "You must
accept terms",
};

```

```

function showError(id, msg) {
const el = qs("#" + id + "-error");
if (!el) return;
el.textContent =
msg === true || msg === false || !msg ?
"" : msg;
}

function validateForm(data, rules) {
let ok = true;
for (const key in rules) {
const rule = rules[key];
const value = data[key];
const res =
rule.length > 1 ? rule(value, data) :
rule(value);
if (res !== true) {
ok = false;
showError(key, res);
} else {
showError(key, "");
} }
return ok;
}

const signupForm = qs("#signup-form");
signupForm.addEventListener("submit", (e)
=> {
e.preventDefault();
const data = {
username: qs("#username").value.trim(),
email: qs("#email").value.trim(),
phone: qs("#phone").value.trim(),

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```
dob: qs("#dob").value,
password: qs("#password").value,
confirmPassword: qs("#confirm-
password").value,
terms: qs("#terms").checked,
};

const rules = {
username: (v) => validators.username(v),
email: (v) => validators.email(v),
phone: (v) => validators.phone(v),
dob: (v) => validators.dob(v),
password: (v) => validators.password(v),
confirmPassword: (v, all) =>
validators.confirmPassword(v, all),
terms: (v) => validators.terms(v),
};

const valid = validateForm(data, rules);
const feedback = qs("#signup-feedback");

if (!valid) {
feedback.textContent = "Please fix errors
above.";
feedback.className = "error";
return;
}

feedback.textContent = "Account created
successfully (demo).";
feedback.className = "success";
signupForm.reset();
});
```

```
const loginForm = qs("#login-form");
loginForm.addEventListener("submit", (e)
=> {
e.preventDefault();

const identifier = qs("#login-
identifier").value.trim();

const password = qs("#login-
password").value;

let ok = true;

if (!identifier) {
showError("login-identifier", "Enter
email or username");
ok = false;
} else {
showError("login-identifier", "");
}

if (!password) {
showError("login-password", "Enter
password");
ok = false;
} else {
showError("login-password", "");
}

const fb = qs("#login-feedback");

if (!ok) {
fb.textContent = "Please fix errors
above.";
fb.className = "error";
return;
}
```

```

}

fb.textContent = "Logged in (demo).";
fb.className = "success";
loginForm.reset();
});

qs("#fill-sample").addEventListener("click", () => {
  qs("#username").value = "demo_user";
  qs("#email").value = "demo@example.com";
  qs("#phone").value = "+12345678901";

  const d = new Date();
  d.setFullYear(d.getFullYear() - 20);

  qs("#dob").value =
    d.toISOString().slice(0, 10);

  qs("#password").value = "Password1";

  qs("#confirm-password").value =
    "Password1";

  qs("#terms").checked = true;
});

qsa("input").forEach((input) => {
  input.addEventListener("input", () => {
    const id = input.id.replace(/-(.+)/, "");
    const err = input.getAttribute("aria-describedby");

    if (id) showError(input.id.replace(/-error$/, ""), "");
  });
});
</script>
</body>
</html>

```

Output:

The screenshot shows a web browser window with the URL `http://localhost:5500/Assignment 1/`. The main content is a form titled "GOONER NIGGAS PORTAL". The form has two tabs: "Login" and "Sign up", with "Sign up" being the active tab. The form fields are as follows:

- USERNAME:** `master_tipu`
- EMAIL:** `tipu05@gmail.com`
- PHONE:** `7779662240`
- DATE OF BIRTH:** `17 / 01 / 2005` (with a calendar icon)
- PASSWORD:** `Qwertyuiop05` (with a "Hide" button)
- CONFIRM PASSWORD:** `.....`
- TERMS AND PRIVACY POLICY:** ☒ I AGREE TO THE TERMS AND PRIVACY POLICY

At the bottom of the form, there are two buttons: "CREATE ACCOUNT" (highlighted in green) and "FILL SAMPLE". Below the buttons, a red error message reads: "Please fix errors above."

GOONER NIGGAS PORTAL

Login Sign up

EMAIL OR USERNAME

master_tipu

PASSWORD

Qwertyuiop05 Hide

Forgot password? Click reset on server.

LOGIN

GOONER NIGGAS PORTAL

Login Sign up

USERNAME

aksjdjnljnl

EMAIL

qnl

Enter a valid email

PHONE

njdjnw

DATE OF BIRTH

dd / mm / yyyy

Enter 10-15 digits, optional +

Date of birth is required

PASSWORD

Min 8 chars with upper, lower and number

CONFIRM PASSWORD

I AGREE TO THE TERMS AND PRIVACY POLICY

You must accept terms

CREATE ACCOUNT FILL SAMPLE

Please fix errors above.

Conclusion:

The program successfully demonstrates data validation by checking user inputs and displaying appropriate error messages before allowing login or account creation.